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Real-Time Camera Soiling Detection & Synthetic Data Generation

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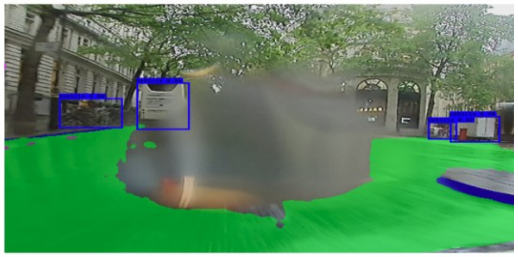
Problem Statement

- ADAS perception degrades under **camera soiling (rain, mud)**
- Labelled real-world soiling data is **scarce and expensive**
- Manual annotation is **not scalable** for production use
- Models trained on clean data **fail to generalise**
- Need for **scalable data generation and robust detection**



Soiling degrades camera perception (example)

Proposed Approach

- **Procedural raindrop synthesis** — physics-based motion, wind decomposition, temporal coherence
- **MudGAN** — mask-guided CycleGAN + VAE for realistic mud generation
- **Paired datasets** — RGB + pixel-level segmentation masks for both soiling types
- **Cross-camera generalisation** — pinhole and fisheye camera geometries

Experimental Design

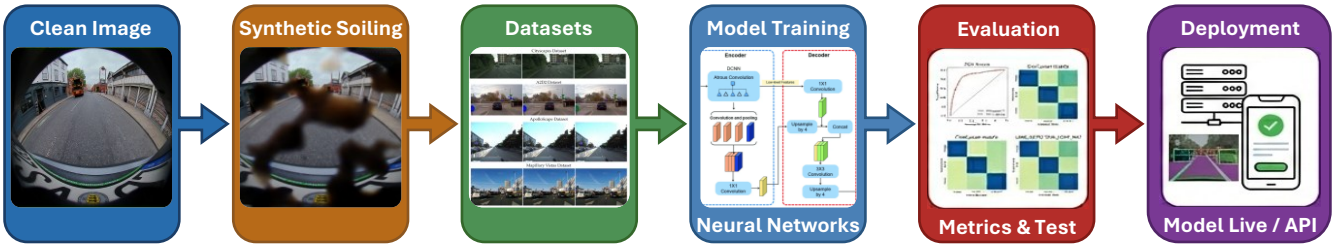
- **Realism evaluation:** FID, KID, LPIPS, MOS scores, Visual Turing Test (n=33)
- **Models benchmarked:** DeepLabV3+, SegFormer, PIDNet
- **4 Training strategies:** *Real only · Synthetic only · Real + Synthetic · Real + Synthetic + Clean priors*

Core Insight: Perceptual Realism ≠ Segmentation Performance

- Synthetic data appears realistic to humans (MOS: 3.87–4.16, ~52–56% misclassified as real)
- Synthetic-only training **fails critically** — Recall drops to 0.17–0.39 despite high Precision (0.98)
- Gap attributed to **missing boundary statistics and transparency cues** in generated data

Key Results

- **Best segmentation:** Dice 0.94 — SegFormer, Real + Synthetic training
- **Best generalisation:** Real + Synthetic + Clean priors — **FPR** reduced to 0.17–1.18% with **high recall**
- **Fisheye outperforms pinhole** by +2–4% Dice across all models
- **PIDNet:** 7 ms inference, 7.7M params — real-time edge deployment ready



Key Takeaway

Effective soiling detection requires real data for grounding, synthetic data for diversity, and clean-image priors for false-positive control — combined training yields **production-ready performance (Dice 0.94, 7 ms inference)**. Synthetic data alone is insufficient for reliable deployment.

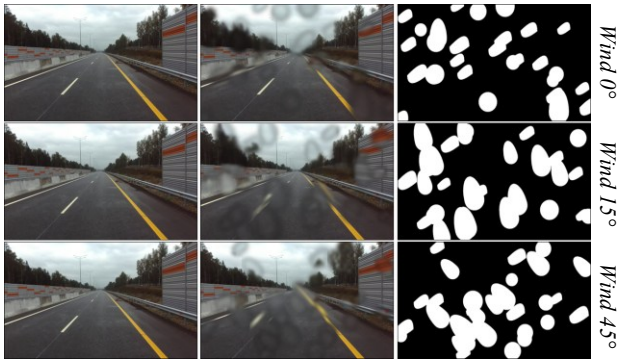
1. Procedural Raindrop Synthesis

Baseline limitations addressed:

- Frame-independent generation → temporal flickering
- No wind/gravity modelling → unrealistic trajectories
- Uniform random placement → no directional rain

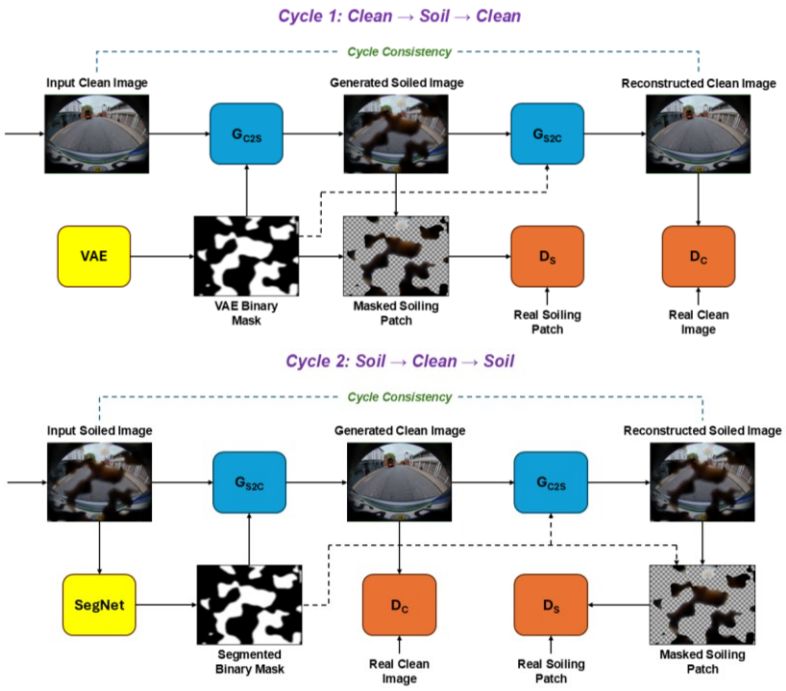
Five enhancements proposed:

- **Physics motion model:** wind decomposition + surface adhesion
- **Temporal state tracking:** persistent droplet IDs, zero flickering
- **Wind-adaptive spawning:** directional rain across 4 regimes
- **Collision physics:** volume-conserving droplet merging
- **Spatial culling:** off-screen drops skipped before rendering

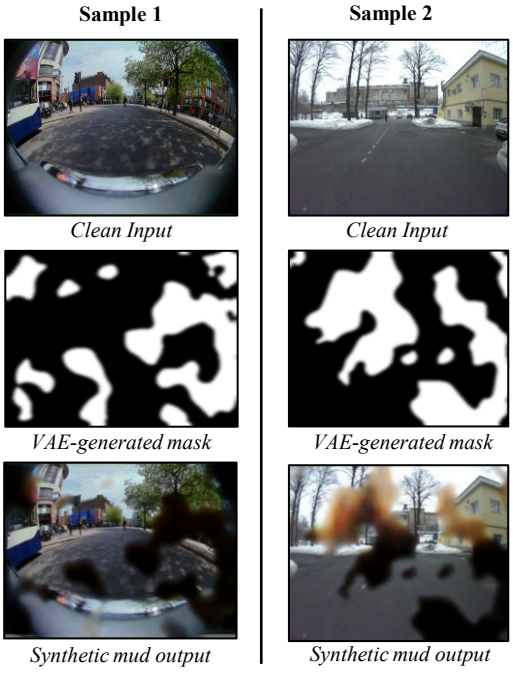


Clean Image *Synthetic Raindrop* *Mask*

2. GAN-Based Mud Synthesis — MudGAN



MudGAN inference



3. Perceptual Evaluation — Quantitative & Human Study

Metric	Raindrops	MudGAN
FID ↓	85.59	113.16
KID* ↓	0.0295 ± 0.006	0.0643 ± 0.011
LPIPS ↓	0.32–0.48	0.32–0.45
MOS ↑ (1–5)	3.87–4.00	3.87–4.16
VTT**	~46%	~56%

**KID preferred over FID due to small reference set sizes (1,000 rain / 2,504 mud images) **Visual Turing Test*

Human Study Summary

- **n=33 participants** — 35 recruited, 2 excluded for unreliable responses
- **MOS avg:** Raindrops 3.93 & MudGAN 3.97 — Good–High per ITU-T scale
- **VTT:** 3/5 synthetic rain images mistaken as real & 5/5 synthetic mud mistaken as real
- **Not Sure responses: 3–30%** — perceptual boundary genuinely unclear to observers

Key Finding: KID score and human perception diverge — MudGAN scores worse on KID (0.064 vs 0.030) but fools more observers in the Visual Turing Test (56% vs 46%). Human observers weight localised texture and material plausibility over global feature statistics — making MOS and VTT more reliable indicators than FID/KID alone.

Model Performance Comparison Table

Model	Case	Dice	IoU	Precision	Recall
DeepLabV3+	Real only	0.91	0.84	0.92	0.91
	Real + Synth + Clean	0.92	0.85	0.92	0.92
SegFormer	Real only	0.93	0.88	0.94	0.93
	Real + Synth	**0.94**	**0.89**	**0.95**	0.93
PIDNet	Real only	0.92	0.85	0.90	**0.94**
	Real + Synth + Clean	0.92	0.86	0.91	**0.94**

Case 2 (synthetic-only) omitted — severe recall failure (0.17–0.39) discussed on Page 1

False Positive Rate + Efficiency

Case	DeepLabV3+	SegFormer	PIDNet
Real only	**4.03%**	2.11%	3.37%
Real + Synth	1.33%	2.39%	0.48%
Real + Synth + Clean	1.18%	0.32%	**0.17%**

Evaluated on 6 clean datasets: urban, highway, parking scenarios

- **PIDNet:** 7 ms | 7.7M params | 29.5 MB — embedded ECU ready
- **SegFormer:** 25 ms | 13.7M params | 52.2 MB — mid-tier GPU optimal
- **DeepLabV3+:** 35 ms | 182.7M params | 174.6 MB — high-performance only

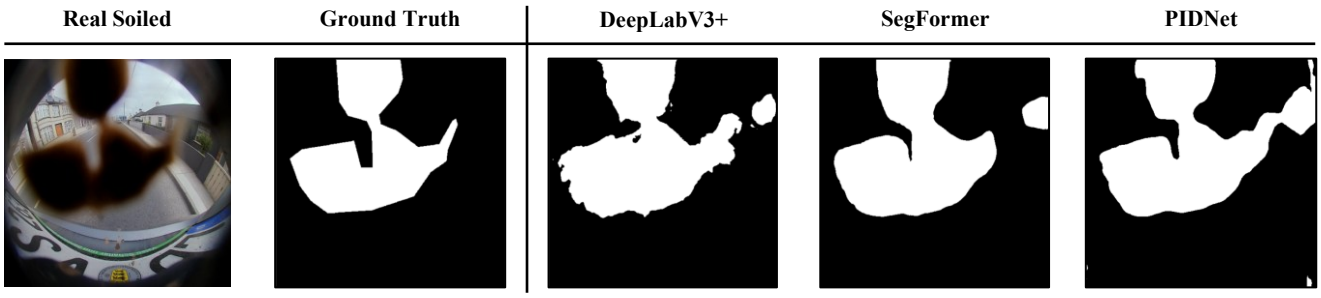
Camera Modality — Fisheye vs Pinhole

- Fisheye outperforms pinhole by +2–4% Dice consistently (e.g. SegFormer Case 3: fisheye 0.95 vs pinhole 0.94)
- Synthetic data aligns more closely with fisheye geometry — largest gap in **Case 2** suggests camera-specific synthesis possible

Deployment Recommendations

- **Embedded ECU (latency-critical):** PIDNet Case 4 — Dice 0.92, FPR 0.17%, 7 ms
- **Mid-tier GPU platform:** SegFormer Case 3 — Dice 0.94, 25 ms
- **High-performance system:** SegFormer preferred over DeepLabV3+ — matched accuracy, lower cost

Why FPR Matters in Deployment: False positives trigger unnecessary camera cleaning — wasting power, depleting onboard water, and causing lens wear from repeated spray and air pressure. Case 4 reduces FPR to **0.17–1.18%** while maintaining Dice **0.92–0.94** — the optimal balance for safe, resource-efficient ADAS deployment.



Case 3 segmentation outputs — Real + Synthetic training across model architectures